

Experimental designs

Sources of Bias

Research ethics

Ethical principles and
current regulations

L10: Randomized designs, bias and ethics

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Learning objectives for today

- ▶ Understanding terminology related to experimental design
- ▶ Spot ways a study can still be biased after randomization
- ▶ Connect trial ethics to the rules used today

Experimental designs

Sources of Bias

Research ethics

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current regulations

Experimental designs

What is an RCT?

You may hear a study referred to as an “RCT” -what is this?

- ▶ R — randomized
- ▶ C — controlled
- ▶ T — trial

Who (or what) receives the assigned treatment?

- ▶ **Individuals:** a patient gets Drug A or Drug B
- ▶ **Households:** a family gets a water filter or usual water
- ▶ **Groups:** a classroom gets a new curriculum or the old one
- ▶ **Clinics:** a clinic starts a reminder system or continues as usual

The unit often becomes a **row** in the data. If we measure the same unit over time, we may have several rows per unit (*long* / longitudinal data).

Discuss: pick one level and invent an exposure that fits it.

Randomization means treatment is assigned **by chance**.

Why bother?

- ▶ People who choose (or are chosen for) treatment are often different at the start
- ▶ Chance tends to balance those starting differences across groups

What does randomization *not* fix by itself?

Music lessons and grades (no randomization)

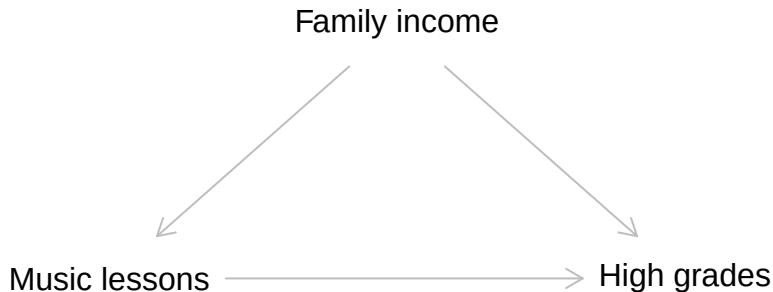
Setting: we sample students and collect information. Some take music lessons; some do not. We then look at grades.

Experimental designs

Sources of Bias

Research ethics

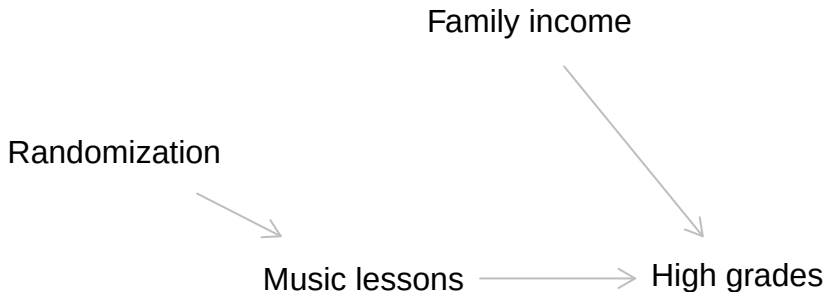
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current regulations



Question: why might lesson-takers have better grades *even if* lessons do nothing?

Music lessons and grades (with randomization)

Setting: eligible students are assigned **by chance** to lessons or no lessons. Family income can still affect grades — but it no longer decides who gets lessons.



Compare the two DAGs: which arrow disappeared?

A simple randomized trial

A “clean” starter picture:

- ▶ the units of randomization are nearly identical
- ▶ only one treatment/exposure group and one control group
- ▶ simple chance assignment
- ▶ effects are immediate or in the short term

Example: fever medicine vs placebo; temperature checked the same evening.

Real trials are often messier!

In every calculation of measure of effect, we must choose a **comparison** group.

Ideal: a **counterfactual** — what would the treated group have looked like under the other exposure?

In a trial we recruit people (or clinics, etc.) and assign them by chance to:

- ▶ intervention, or
- ▶ control (usual care, older drug, placebo, ...)

Discuss: why not one treated person vs one control person?

Setting: 50,000 people all get a vaccine. No one is held back for comparison. In the next 3 weeks, 0.1% report an adverse event.

Does this show the vaccine is harmful?

What is missing?

- ▶ We do not know the **background rate** of the same events among similar people who did **not** get the vaccine

Other common trial terms

- ▶ **Cluster randomized:** assign whole groups together (schools, clinics)
- ▶ **Community randomized:** assign whole communities
- ▶ **Blind / double / triple blind:** who does *not* know the assignment
- ▶ **Placebo controlled:** comparison arm uses an inactive look-alike

For each label, ask: who got what, and who knew?

Blinding



Blinding = keep the treatment assignment hidden from someone.

Why? Knowing the arm can change what people report, look for, or score.

Levels of blinding

- ▶ **Single blind:** usually participants
- ▶ **Double blind:** participants + people scoring outcomes
- ▶ **Triple blind:** often also investigators / analysts

(Definitions vary — read the methods.)

When is full blinding hard? Surgery, exercise, counseling, cooking classes. . .

Then what can we still do?

Placebo



A **placebo** looks like treatment but has no active ingredient.

It helps with blinding, and it helps separate drug effects from expectations and clinic care.

When is a pure placebo arm a problem?

What is the placebo effect?

People can feel better after inactive treatment. Reasons can include expectation, care, and time (recovery on its own).

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Research ethics

Ethical principles and
current regulations



[NYT Magazine](#), Nov 2018

Trial reading tip: both arms may improve. Look at the **difference**.

Why not always randomize?

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Research ethics

Ethical principles and
current regulations

If RCTs are so strong, why not learn *only* from them?

- ▶ Who joins / who stays may be a selected group
- ▶ Cost, logistics, feasibility
- ▶ not a good measure of real world effects
- ▶ not possible or ethical to randomize
- ▶ time needed to get an answer

and others

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Sources of Bias

Research ethics

Ethical principles and
current regulations

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Randomization balances arms **inside the enrolled sample**.

It does not fix a sampling frame that missed part of the target population (**under-coverage** bias).

Example: a depression drug trial recruits only from specialty clinics. Who never had a chance to enter?

1. Non-response: individual decline to participant

- ▶ If there is enough information about who did and did not respond, the data may be adjusted to try and compensate for this type of bias. This kind of adjustment is common in survey samples.

2. Lost to follow-up: participants drop out or become lost to follow-up

- ▶ If one of your study groups is more likely to be lost this can create bias.

(sicker people)

Contamination: When the study participants randomized to control receive the medication or exposure that was intended to be given only to treatment arm participants.

Example setting (lifestyle): control participants hear the study goals, then change diet/exercise on their own. Arms look more alike. The estimated effect can shrink toward **no difference**.

Related cases: MRFIT (heart risk counseling) and SHAZ! (community HIV program).

Once a participant is randomized to a given treatment, they do not always adhere to that treatment.

Two common analyses:

- ▶ **Intention to treat (ITT)**: keep people in the arm they were **assigned**
 - ▶ answers: “what happened when we *offered* this plan?”
- ▶ **Per protocol**: analyze by what people **actually took**
 - ▶ can describe adherers — but adherers are a selected group

Randomization does not fix bad measurement.

- ▶ imprecise measurement equipment
- ▶ poorly designed or worded questions
- ▶ responses of the individual to the measurement process
 - ▶ white-coat hypertension: BP rises in clinic
 - ▶ sleep lab: sleep may change just because you are in a lab

Random error adds noise. Systematic error pushes results in one direction — and big n will not cancel it.

People may not report (or remember) the full story:

- ▶ **social desirability**: answers that sound “good”
- ▶ **poor recall**: forgetting
- ▶ **biased recall**: a bad outcome changes how people search memory
- ▶ **diary non-compliance**: filled in later from a blurry week

Small samples are noisy (think: eight online reviews).

Often trials are designed to find a **statistically significant** difference between treatment and control. This means an observed effect so large that it would rarely occur by chance.

Size helps **precision**. It does not wash away **bias**.

Experimental designs

Sources of Bias

Research ethics

Ethical principles and
current regulations

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This is a methods course — and the “units” are often people.

Next: a few historical cases that help explain today’s rules (consent, IRBs, equipoise).

Is randomization ethical (when it is possible)?

Possible $\neq$ okay to do.

Is it ethical to randomize people to:

- ▶ lack of sleep?
- ▶ exposure to an infectious disease?
- ▶ fetal exposure via maternal smoking or alcohol consumption?

Ethical principles and current regulations

Key influential documents: Nuremberg

After WWII, the Nuremberg Code answered medical crimes by Nazi doctors.

It feeds into U.S. rules (Title 45/46 CFR) and IRB review. Human subjects research in the U.S. generally needs IRB approval first.

Nuremberg principles (1–3)

1. The voluntary consent of the human subject is absolutely essential.
2. The experiment should be such as to yield fruitful results for the good of society, unprocurable by other methods or means of study, and not random and unnecessary in nature.
3. The experiment should be so designed and based on the results of animal experimentation and a knowledge of the natural history of the disease or other problem under study that the anticipated results will justify the performance of the experiment.

Nuremberg principles (4–7)

4. The experiment should be so conducted as to avoid all unnecessary physical and mental suffering and injury.
5. No experiment should be conducted where there is an a priori reason to believe that death or disabling injury will occur; except, perhaps, in those experiments where the experimental physicians also serve as subjects.
6. The degree of risk to be taken should never exceed that determined by the humanitarian importance of the problem to be solved by the experiment.
7. Proper preparations should be made and adequate facilities provided to protect the experimental subject against even remote possibilities of injury, disability, or death.

Nuremberg principles (8–10)

8. The experiment should be conducted only by scientifically qualified persons. The highest degree of skill and care should be required through all stages of the experiment of those who conduct or engage in the experiment.
9. During the course of the experiment the human subject should be at liberty to bring the experiment to an end if he has reached the physical or mental state where continuation of the experiment seems to him to be impossible.
10. During the course of the experiment the scientist in charge must be prepared to terminate the experiment at any stage, if he has probable cause to believe, in the exercise of the good faith, superior skill and careful judgment required of him that a continuation of the experiment is likely to result in injury, disability, or death to the experimental subject.

Key influential documents: Belmont

The Belmont Report (1978) followed scandals including Tuskegee.

Three lenses used in U.S. review:

- ▶ respect for persons (autonomy + protection)
- ▶ beneficence (help more / harm less)
- ▶ justice (fair sharing of risks and benefits)

Consent needs more than a signature.

- ▶ What must people know? (purpose, risks, benefits, alternatives, right to leave)
- ▶ Who can consent? (adults; parents/guardians for children + assent when possible)

Can the person **understand** the form — and freely say no?

Even with a signed form, “yes” can be bent:

- ▶ payments that are hard to refuse
- ▶ study as the only path to scarce care
- ▶ fear that saying no hurts ordinary medical care

Useful test: *given this person's situation, could they decline?*

Equipoise: experts are genuinely unsure which arm is better.

- ▶ at the start, neither arm should be clearly known to be worse
- ▶ if one arm is already believed worse/more harmful — lottery is hard to justify

TCPS 2 reference

Who benefits, who is included?

There is also a larger issue of justice and ethics related to who benefits from the trial.

- ▶ Is there a direct benefit to participants?
- ▶ Is there a potential benefit to individuals who are similar to the study participants?
- ▶ Is there a group that is excluded from participation unfairly?

- ▶ Maintaining anonymity of participants is more complex in an era where data are collected constantly
- ▶ Removing a name or id number is no longer sufficient to protect the identity of a participant
- ▶ Push to share data and/or make data publically available

For Indigenous communities, data have often been collected, stored, and interpreted by outside institutions.

Ask: who owns the file? who approves reuse? who tells the story?

- ▶ FNIGC (2019); Kukutai & Taylor (eds., 2016)

Principles of Indigenous data sovereignty

L10: Randomized designs, bias and ethics

Experimental designs

Sources of Bias

Research ethics

Ethical principles and current regulations

- ▶ extraction has reinforced unequal power
- ▶ framing has emphasized deficit / stereotypes
- ▶ state data are often of little use to Nations
- ▶ governments assumed ownership by default
- ▶ Nation-to-Nation governance conversations have been thin

Source: BCFNDGI / Open North, *Decolonizing Data* (2017)

Parting humor



**"Miss Peterson, may I go home? I can't assimilate
any more data today."**